QUIZ 6

Instructions: Please write your name at the top of the page. Circle T if the statement is always true under the stated hypotheses, and F otherwise. No justification is required. Each question is worth 1 point (10 points total).

Unless otherwise indicated, $\mathbb{K}$ is a field and all vector spaces are finite-dimensional over $\mathbb{K}$.

(a) **T** **F** If $V \neq 0$, then $T(V)$ is infinite-dimensional.

(b) **T** **F** The map $V \to V^{\otimes 2}$ given by $v \mapsto v \otimes v$ is linear.

(c) **T** **F** If $\dim(V) = 3$, then $\dim(\mathrm{Sym}^2(V)) = 9$.

(d) **T** **F** If the columns of an invertible matrix P express a new basis in terms of the old basis, then a Gram matrix G changes to $P^{-1}GP$.

(e) **T** **F** Every alternating bilinear form on an odd-dimensional vector space is degenerate, including in characteristic 2.

(f) **T** **F** The determinant pairing induces a natural isomorphism $\Lambda^d(V^\vee) \cong \left(\Lambda^d(V)\right)^\vee$ for every $d \geq 0$, including in characteristic 2.

(g) **T** **F** Two symmetric bilinear forms on $\mathbb{R}^n$ with the same rank are isometric.

(h) **T** **F** In $\Lambda^3(\mathbb{Q}^3)$, we have

$$(1,1,0) \wedge (0,1,1) \wedge (1,2,1) = 0.$$

(i) **T** **F** If e_1, e_2, e_3 is the standard basis of $\mathbb{Q}^3$, then

$$(e_1 + 2e_2) \wedge (e_2 + e_3) = e_1 \wedge e_2 + e_1 \wedge e_3 - 2e_2 \wedge e_3.$$

(j) **T** **F** Let e_1, e_2 be the standard basis of $\mathbb{Q}^2$, and let $\phi : \mathbb{Q}^2 \to \mathbb{Q}^2$ be the linear map defined by $\phi(e_1) = e_1 + e_2$ and $\phi(e_2) = e_1 - e_2$. Then

$$\mathrm{Sym}^2(\phi)(e_1 e_2) = e_1^2 - e_2^2.$$

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